

OCR B A-Level Biology — Practical Activities

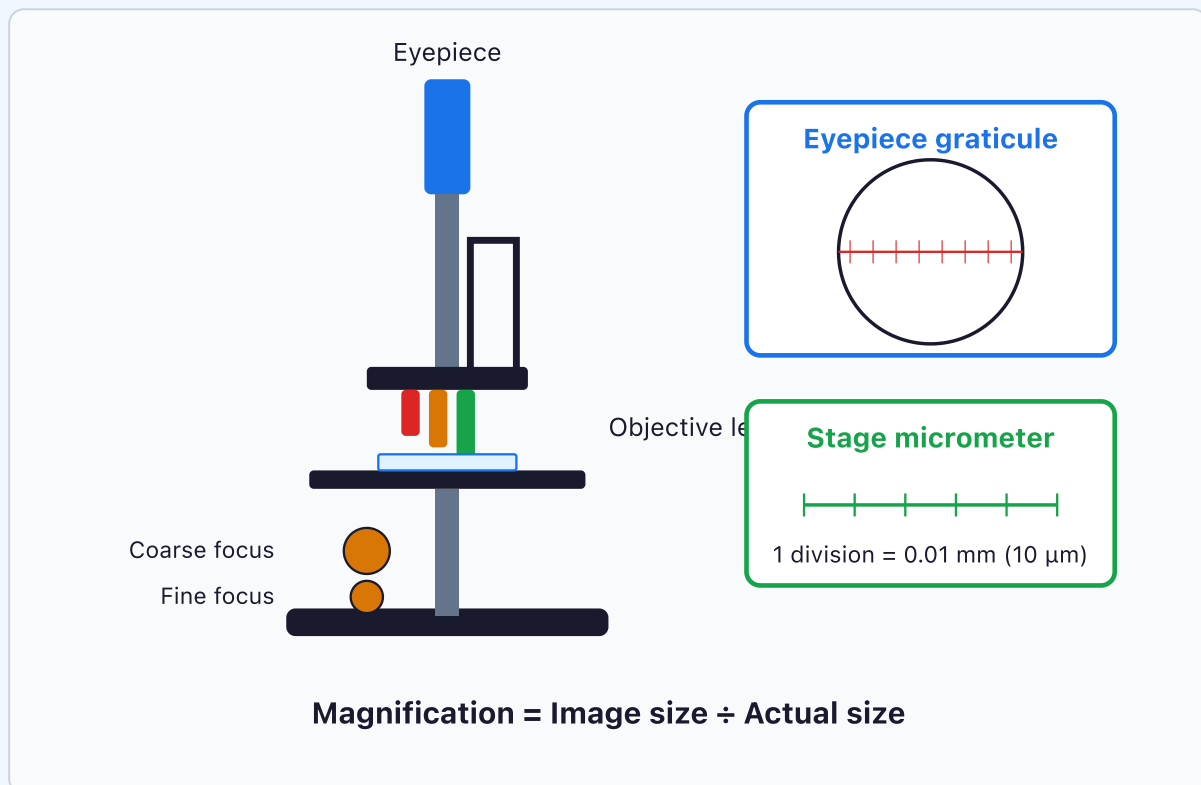
The OCR A-Level Biology B (H422) Practical Endorsement is organised into twelve **Practical Activity Groups (PAGs)**: PAG1 Microscopy, PAG2 Dissection, PAG3 Sampling techniques, PAG4 Rates of enzyme-controlled reactions, PAG5 Colorimeter or potometer, PAG6 Chromatography or electrophoresis, PAG7 Microbiological techniques, PAG8 Transport in and out of cells, PAG9 Qualitative testing, PAG10 Investigation using a data logger or computer modelling, PAG11 Investigation into the measurement of plant or animal responses, and PAG12 Research skills. You must complete at least twelve practical activities covering all twelve groups. Schools can choose their own activities, so the practicals below are typical examples for each group.

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PAG1: Microscopy — Measuring Cells with a Calibrated Graticule



Background Theory

Light microscopy allows the observation of cells and tissues at magnifications up to ~1500x with a maximum resolution of approximately **200 nm** (limited by the wavelength of visible light, 400–700 nm). Resolution is the minimum distance between two distinguishable points — more important than magnification for revealing fine detail. The **total magnification** = eyepiece magnification x objective magnification (e.g. x10 eyepiece x x40 objective = x400 total).

To measure actual cell sizes, an **eyepiece graticule** (a ruler inside the eyepiece) must be **calibrated** against a **stage micrometer** (a slide with known graduations, typically 0.01 mm = 10 µm per division). Calibration must be repeated for each objective lens because the scale changes with magnification.

Apparatus

- Light microscope with objective lenses (x4 low power, x10 medium, x40 high power)
- Eyepiece graticule (fitted inside the eyepiece lens)
- Stage micrometer (calibration slide with known scale markings)
- Glass slides, coverslips, mounted needles, forceps, pipette
- Stains: methylene blue (nuclei/animal cells), iodine in potassium iodide (starch/plant cells), eosin (cytoplasm), aceto-orcein/toluidine blue (chromosomes)
- Specimens: onion epidermis, cheek epithelial cells, prepared slides (blood smear, root tip sections)

Method

1. **Calibrate the eyepiece graticule:** Place the stage micrometer on the stage. Align the eyepiece graticule scale with the stage micrometer scale. Count how many eyepiece divisions correspond to a known number of stage micrometer divisions. Calculate: value of one eyepiece division = (number of stage divisions aligned x value of each stage division) / number of eyepiece divisions.
2. **Prepare a wet mount:** Place a thin specimen on a clean glass slide. Add a drop of water or appropriate stain. Lower a coverslip at 45 degrees using a mounted needle to exclude air bubbles. Blot excess liquid with filter paper.
3. **Focus:** Start on the **lowest power** objective (x4). Use the **coarse focus** knob to bring the specimen into rough focus, then switch to **fine focus** for sharpness. Move to higher magnifications, refocusing with fine focus only (to avoid crushing the slide).
4. **Measure specimens:** Count how many calibrated eyepiece divisions the specimen spans. Multiply by the value of each division to obtain the actual size.
5. **Produce biological drawings:** Use a sharp pencil. Draw clear, continuous lines (no sketching or shading). Ensure proportions are accurate. Label with ruled horizontal lines and neat annotations. Include a title, magnification, and scale bar.

Key Calculations

Magnification = Image size / Actual size ($M = I / A$)

Rearranged: Actual size = Image size / Magnification; Image size = Actual size x

Magnification

Unit conversions: 1 mm = 1,000 μm ; 1 μm = 1,000 nm

Always ensure image size and actual size are in the **same units** before calculating.

Calibration example: If 50 eyepiece graticule divisions align with 10 stage micrometer divisions, and each stage division = 10 μm , then one eyepiece division = $(10 \times 10) / 50 = 2 \mu\text{m}$. If a cell spans 30 eyepiece divisions, its actual width = $30 \times 2 = 60 \mu\text{m}$.

Safety

- Handle glass slides and coverslips carefully (sharp edges can cut).
- Methylene blue and iodine stain skin and clothing — wear gloves and lab coat.
- If using cheek cells (human tissue): sterilise spatula by dipping in disinfectant before and after scraping; dispose of slides into 1% Virkon disinfectant solution.
- Carry microscopes with two hands (one on the arm, one under the base).

Sources of Error and Improvements

Error	Effect	Improvement
Parallax error when aligning graticule with stage micrometer	Miscalibration; all subsequent measurements systematically wrong	Close one eye; ensure scales are precisely aligned; repeat calibration 3 times and take the mean
Forgetting to recalibrate when changing objective lens	Measurements are wrong by a large factor (systematic error)	Always recalibrate the eyepiece graticule each time you change the objective magnification
Air bubbles trapped under the coverslip	Distort the image; can be mistaken for cells or structures	Lower the coverslip slowly at 45 degrees; use a mounted needle to guide it down; tap gently to release bubbles
Cells not in a single layer (overlapping)	Difficult to measure individual cells; confusing images	Tease tissue apart with mounted needles; use very thin sections; squash preparations gently

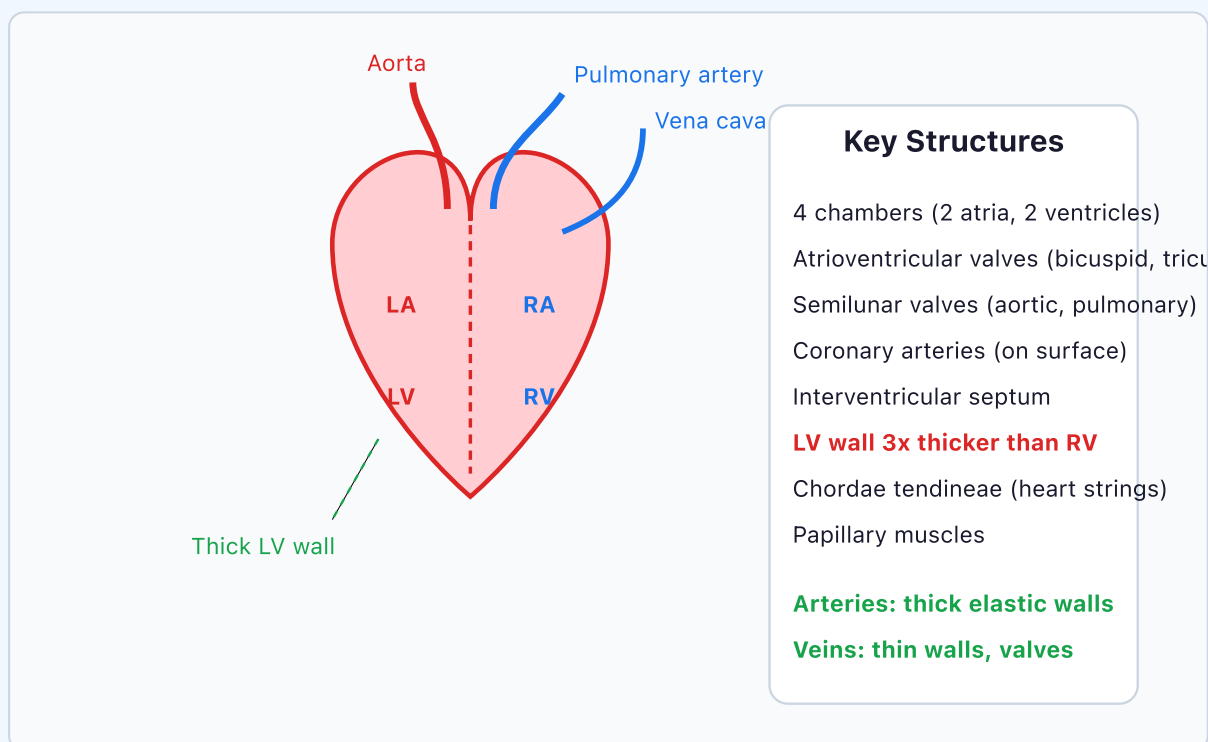
Image size measured inaccurately in drawings

Magnification calculation is wrong

Use a ruler to measure drawings precisely in mm; show working clearly

Exam Tip: Always show working for magnification calculations: state the formula ($M = I / A$), convert units so both are the same, then calculate. Include the magnification on all biological drawings (e.g. "x400"). If given a photomicrograph with a scale bar, measure the scale bar with a ruler and use it to calculate the actual size of structures in the image. A very common exam question gives you an image and a scale bar, then asks for the actual size of a cell or organelle.

PAG2: Dissection — the Mammalian Heart



Background Theory

Dissection of organs allows direct observation of the relationship between **structure and function**. The mammalian heart is a **double pump**: the right side pumps deoxygenated blood to the lungs (pulmonary circulation) at lower pressure, while the left side pumps oxygenated blood

to the body (systemic circulation) at higher pressure. This explains why the **left ventricle wall is approximately three times thicker** than the right — it must generate much higher pressure to push blood around the entire body. The **complete separation** of oxygenated and deoxygenated blood (four-chambered heart with septum) maximises the efficiency of oxygen delivery to tissues, supporting the high metabolic rate of endotherms.

Apparatus

- Fresh heart specimen (lamb or pig, from a butcher), or lung/kidney specimen
- Dissecting board, scalpel (sharp blade), scissors (fine and blunt-ended), forceps, mounted needle, seeker (blunt probe)
- Dissecting pins, paper towels
- Disposable gloves, safety goggles, lab coat
- Biological waste container

Method (Heart Dissection)

1. Place the heart on the board with the **anterior surface facing you**. Identify the front by locating the coronary arteries (visible on the surface) and the interventricular groove.
2. Identify the major vessels: **aorta** (thick-walled, elastic — feel it), **pulmonary artery** (in front of aorta, thinner walls), **vena cava** (thin-walled, posterior), **pulmonary veins** (entering the left atrium from behind).
3. Insert a blunt seeker or finger into each vessel to determine which chamber it connects to. This confirms identification.
4. Using scissors, make a **longitudinal cut** from the pulmonary artery down through the right ventricle wall. Open the heart to reveal the right atrium and ventricle. Identify the **tricuspid valve** (3 flaps), **chordae tendineae** (tendinous cords connecting valve flaps to papillary muscles), and the **pulmonary semilunar valve**.
5. Make a second longitudinal cut through the **left side** from the aorta through the left ventricle. Identify the **bicuspid (mitral) valve** (2 flaps), chordae tendineae, and **aortic semilunar valve**.
6. Compare the **wall thickness** of the left ventricle vs the right ventricle vs the atria. Measure with a ruler.
7. Identify the **interventricular septum** separating left and right sides.

8. Produce **labelled biological drawings** of external and internal views, with annotations linking structure to function.

Structure-Function Relationships

Structure	Function
Thick muscular left ventricle wall (~12–15 mm)	Generates high pressure to pump blood through the systemic circulation (entire body). Must overcome greater resistance
Thinner right ventricle wall (~3–5 mm)	Only pumps blood to nearby lungs (short distance); lower pressure prevents damage to delicate pulmonary capillaries
Very thin atrial walls	Atria only push blood a short distance into the ventricles below; do not need thick muscle
Atrioventricular valves (bicuspid and tricuspid) with chordae tendineae	Prevent backflow of blood from ventricles to atria during ventricular systole. Chordae tendineae prevent valves inverting under pressure
Semilunar valves (aortic and pulmonary)	Prevent backflow from arteries back into ventricles during ventricular diastole
Coronary arteries (on heart surface)	Supply the heart muscle itself with oxygenated blood for its own aerobic respiration. Blockage causes myocardial infarction (heart attack)
Interventricular septum	Completely separates oxygenated (left) and deoxygenated (right) blood, maintaining efficient oxygen delivery

Safety

- Wear **disposable gloves** at all times when handling biological specimens (risk of bacterial contamination, including *Salmonella* and *Campylobacter*).
- Wear **safety goggles** and a lab coat to protect eyes and clothing.

- Use sharp instruments carefully; always **cut away from the body** and keep fingers clear of the cutting line.
- Dispose of all biological material in designated biohazard waste containers for autoclaving or incineration.
- Clean work surfaces with disinfectant after the practical. Wash hands thoroughly with soap and warm water.

Sources of Error and Improvements

Error	Effect	Improvement
Difficulty identifying vessels (especially if trimmed by butcher)	Structures misidentified; incorrect labelling	Insert a seeker or finger into each vessel to find which chamber it connects to; look up reference diagrams beforehand
Cutting too deep or in the wrong plane	Damages internal structures (valves, chordae tendineae)	Use scissors rather than scalpel for opening; cut slowly and carefully; use blunt-ended scissors
Tissue drying out during prolonged examination	Structures shrink and become harder to identify	Keep the specimen moist with paper towels dampened with water
Biological drawings not to scale or poorly labelled	Loses marks in assessment	Use pencil with continuous lines; maintain accurate proportions; use ruled label lines; include magnification

Exam Tip: Always link structure to function in dissection questions. For the heart: "The left ventricle wall is thicker than the right because it must generate higher pressure to pump blood through the systemic circulation (around the entire body), whereas the right ventricle only pumps blood the short distance to the lungs." For arteries vs veins: arteries have thick elastic walls to withstand and smooth out high-pressure pulsatile flow; veins have thinner walls and valves to prevent backflow of low-pressure blood.

PAG3: Sampling Techniques — Abundance and Species Diversity

Background Theory

Ecologists estimate population sizes and species diversity using sampling techniques because counting every organism is impractical. The two main approaches are **quadrat sampling** (for sessile organisms like plants) and **mark-release-recapture** (Lincoln index, for mobile animals). **Random sampling** eliminates selection bias: use random number tables or a calculator to generate coordinates for quadrat placement. Transect sampling (belt or line transects) is used when investigating changes across an environmental gradient (e.g. from shoreline to inland, or from woodland edge to interior).

Apparatus

- Quadrats (0.25 m^2 or 1 m^2), tape measures (2, laid at right angles), random number table/calculator
- Pitfall traps (for ground-dwelling invertebrates), sweep nets (for meadow invertebrates), pooters, pond nets
- Non-toxic marker (waterproof paint, correction fluid) for mark-release-recapture
- Species identification keys, field guides, tally charts, recording sheets

Methods

A. Random Quadrat Sampling:

1. Lay two tape measures at right angles along two edges of the study area to form a coordinate grid.
2. Generate pairs of random numbers (x, y coordinates) using a random number table or calculator.
3. Place the quadrat at each random coordinate. Record: (a) number of individuals of each species (for countable organisms), (b) **percentage cover** (for spreading species like grasses — estimate what proportion of the quadrat area is covered), or (c) **frequency** (presence/absence in each quadrat).

4. Use at least 10–30 quadrats (more for heterogeneous habitats). Calculate mean density per quadrat, then scale up to the total area to estimate population size.

B. Belt Transect (for environmental gradients):

1. Lay a tape measure along the line of the environmental gradient.
2. Place quadrats at regular intervals (e.g. every 2 m) along the transect and record species composition in each.
3. Also measure the abiotic factor changing along the gradient (e.g. light intensity, soil moisture, altitude).
4. Plot species distribution against the abiotic variable.

C. Mark-Release-Recapture (Lincoln Index):

1. **Capture** a sample of animals using an appropriate method. Count them: this is **M** (total marked).
2. **Mark** each individual with a small, non-toxic, inconspicuous mark. The mark must not affect survival, predation vulnerability, or behaviour.
3. **Release** all marked individuals back into the habitat. Allow enough time for random mixing with the unmarked population (typically 24+ hours for invertebrates).
4. **Recapture** a second sample. Count the total in this second sample: **n**. Count how many in this sample are already marked: **m**.
5. Estimate total population: **$N = (M \times n) / m$**

Key Formulae

Lincoln Index: $N = (M \times n) / m$

Where: N = estimated population size, M = number marked and released, n = total number in second sample, m = number of marked individuals recaptured.

Simpson's Diversity Index: $D = 1 - \sum(n/N)^2$

Where n = number of individuals of each species, N = total number of all individuals. D ranges from 0 (no diversity) to 1 (maximum diversity). Higher D = more biodiverse habitat.

Lincoln Index Assumptions

- Marks do not affect survival, predation risk, or behaviour
- Marks do not rub off or fade between capture events
- Population is **closed**: no immigration, emigration, births, or deaths between samples
- Marked individuals mix randomly with unmarked individuals before recapture
- Both capture methods have equal probability of capturing marked and unmarked individuals

Safety

- Complete a **fieldwork risk assessment** (weather, terrain, water hazards, stinging plants, tick exposure, allergies)
- Handle organisms gently and return to habitat promptly
- Wash hands thoroughly after handling soil, water, or organisms
- Wear appropriate clothing, sturdy footwear, and sun protection

Sources of Error and Improvements

Error	Effect	Improvement
Non-random quadrat placement (avoiding difficult terrain)	Biased sample; species counts not representative	Use random number table strictly; accept whatever falls in the quadrat
Marks wear off between capture events	Underestimates m (recaptured marked); overestimates N	Use permanent, waterproof markers; test mark durability before the experiment
Marked animals more visible to predators	Marked individuals preferentially eaten; m decreases; N overestimated	Use small, inconspicuous marks placed where predators cannot see them (e.g. underside)

Immigration or emigration between sampling events	Population not closed; N inaccurate (immigration of unmarked or emigration of marked animals lowers the marked proportion, so N is overestimated)	Keep the interval between captures short; use a clearly bounded habitat
Insufficient number of quadrats	High variance; mean not representative of the true population	Use enough quadrats that the running mean stabilises (at least 10, ideally 20+)

Exam Tip: Lincoln index: $N = (M \times n) / m$. Always state and explain the assumptions for full marks. If marks affect survival (e.g. make organisms more conspicuous to predators), fewer marked individuals are recaptured, m decreases, and N is **overestimated**. For Simpson's D: calculate step by step — find n/N for each species, square each value, sum all the squared values, then subtract from 1. A D close to 1 indicates high biodiversity (many species in relatively equal abundance); D close to 0 indicates low biodiversity (dominated by one species).

PAG4: Rates of Enzyme-Controlled Reactions — Enzyme Activity

Background Theory

Enzyme activity is affected by **temperature**, **pH**, **substrate concentration**, **enzyme concentration**, and the presence of **inhibitors**. This practical typically investigates one or more of these factors using catalase ($\text{H}_2\text{O}_2 \rightarrow \text{H}_2\text{O} + \text{O}_2$) or amylase (starch $\rightarrow$ maltose). The key concepts tested are: the relationship between kinetic energy and collision rate, enzyme denaturation at extreme temperatures/pH, enzyme saturation at high substrate concentrations (V_{max}), and competitive vs non-competitive inhibition.

OCR's own example for PAG4 is the effect of **substrate concentration** on the rate of an enzyme-controlled reaction. You should be able to find the **initial rate** (gradient of a tangent to

the product/time curve at $t=0$). Q_{10} is a useful way to compare rates at temperatures 10°C apart, but it is not named in the H422 specification.

Key Methods Summary

Investigation	Enzyme/Substrate	How to Measure Rate	Expected Result
Effect of temperature	Amylase + starch, or catalase + H_2O_2	Amylase: time for iodine to stop turning blue (rate = $1/t$). Catalase: volume of O_2 per minute (gas syringe)	Bell-shaped curve; rate increases to optimum then falls sharply as enzyme denatures
Effect of pH	Amylase + starch in buffer solutions (pH 2–10)	Time for starch digestion (iodine test); rate = $1/t$	Bell-shaped curve; optimum pH specific to enzyme (amylase $\sim$ pH 7, pepsin $\sim$ pH 2)
Effect of substrate concentration	Catalase + varying $[\text{H}_2\text{O}_2]$	Initial rate of O_2 production (tangent at $t=0$)	Hyperbolic curve: rate increases then plateaus at V_{max} (enzyme saturation)
Effect of enzyme concentration	Catalase (varying mass of potato) + fixed $[\text{H}_2\text{O}_2]$	Initial rate of O_2 production	Linear increase (more active sites available); assumes excess substrate

Critical Practical Point: Equilibrate Separately

Always equilibrate enzyme and substrate **in separate test tubes** at the target temperature/pH for **at least 5 minutes** before mixing. If you mix them first and then heat, the initial reaction occurs at the wrong temperature, and your rate data is invalid.

Sources of Error and Improvements

Error	Effect	Improvement
Gas escapes before bung is fitted (catalase experiments)	Underestimates the volume of O ₂ produced, especially at high substrate concentrations where the reaction is fastest	Use a side-arm flask with a septum; inject substrate through the septum to start the reaction without opening. Practise fast bung fitting
Not equilibrating enzyme and substrate before mixing	Initial rate does not reflect the target temperature; invalidates comparisons	Equilibrate both tubes separately for 5 minutes before mixing. Use a thermostatically controlled water bath
Subjective end-point judgement (iodine test)	Inconsistent determination of when starch is fully digested	Use a colorimeter set to the appropriate wavelength for an objective, quantitative measurement
Using total product at a fixed time instead of initial rate	At later time points, substrate is depleted and rate has slowed; does not reflect the rate at the nominal concentration	Calculate initial rate from the gradient of the tangent to the curve at $t = 0$

Exam Tip: Rate = $1/\text{time}$ for amylase experiments. For catalase, rate = volume of O₂ produced / time (from gradient of tangent at $t=0$). Always use **initial rate** when comparing different conditions. $Q_{10} = \text{rate at } (T+10) / \text{rate at } T$; a Q_{10} of ~ 2 confirms the process is enzyme-catalysed. Denaturation is **permanent and irreversible** — do not say the enzyme is "killed" (enzymes are not alive) or that it will recover when cooled.

PAG4: Rates of Enzyme-Controlled Reactions — Photosynthesis

Background Theory

This practical investigates limiting factors on the rate of photosynthesis, typically using aquatic pondweed (*Elodea* or *Cabomba*) that produces visible O₂ bubbles. **Blackman's law of limiting factors** states that at any given time, the rate of a physiological process is limited by the factor in shortest supply. For photosynthesis, the three main limiting factors are **light intensity**, **CO₂ concentration**, and **temperature**.

The standard method counts O₂ bubbles (or collects the gas) from pondweed in sodium hydrogencarbonate solution, with a lamp at different distances. The OCR B specification lists light wavelength, light intensity, light duration, temperature and pH as factors to investigate, and also expects practical investigation of the **Hill reaction**: isolated chloroplasts in the light reduce the blue dye **DCPIP**, which acts as an artificial electron acceptor and turns colourless, and the rate can be followed with a colorimeter. The **leaf disc flotation method** below is a useful alternative to bubble counting.

Leaf Disc Flotation Method (Alternative to Bubble Counting)

1. Use a single-hole punch to cut uniform leaf discs from a fresh leaf (e.g. spinach). Punch ~10 discs per treatment.
2. Place the discs in a syringe containing NaHCO₃ solution. Draw back the plunger to create a partial vacuum — this removes air from the intercellular spaces, causing the discs to **sink**.
3. Transfer the sunken discs to beakers of NaHCO₃ solution at different distances from a light source.
4. As photosynthesis occurs, O₂ produced fills the intercellular spaces, making the discs buoyant. They **float to the surface**.
5. Record the **time for each disc to float**, or count the number of floating discs at regular intervals (e.g. every minute). Rate of photosynthesis = 1 / time to float.

Key Points

- NaHCO₃ provides a constant supply of dissolved CO₂ so it does not become limiting.
- Light intensity follows the inverse square law: $I \propto 1/d^2$.
- A heat shield (water-filled beaker) between lamp and plant prevents temperature confounding.

- The rate initially increases linearly with light intensity (light-limited), then plateaus (CO₂ or temperature becomes limiting).

Exam Tip: The leaf disc method is more quantitative than bubble counting because every disc is the same size. Rate = $1/t$ (where t = time for disc to float). For the highest marks, explain why the plateau occurs: "At high light intensities, the rate of the light-dependent reactions exceeds the capacity of the Calvin cycle (limited by CO₂ fixation by RuBisCO or the regeneration of RuBP), so another factor has become limiting." Always mention that NaHCO₃ is added to prevent CO₂ becoming limiting.

PAG4: Rates of Enzyme-Controlled Reactions — Respiration

Background Theory

This practical uses a **respirometer** to measure the rate of O₂ consumption by germinating seeds (or small invertebrates) at different temperatures. The organisms are sealed in a tube with soda lime, which absorbs the CO₂ they release; as they take up O₂ the gas volume falls and a coloured liquid drop in the capillary tube moves towards them. The distance moved per minute, multiplied by the cross-sectional area of the capillary (πr^2), gives the volume of O₂ used per minute. The OCR B specification covers how temperature, substrate concentration, anaerobic conditions and the type of respiratory substrate affect respiration rate, and the **respiratory quotient (RQ)**:

RQ = CO₂ produced / O₂ consumed

RQ for glucose (aerobic): $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O \Rightarrow RQ = 6/6 = 1.0$

RQ for lipid (e.g. tripalmitin): $C_{51}H_{98}O_6 + 72.5O_2 \rightarrow 51CO_2 + 49H_2O \Rightarrow RQ = 51/72.5 = 0.7$

RQ for protein: approximately **0.8** (varies with amino acid composition)

RQ > 1.0 indicates anaerobic respiration is occurring (CO₂ produced without O₂ consumption)

Measuring RQ Experimentally

To determine RQ, you need two respirometer setups:

1. **With soda lime:** Measures O_2 uptake only (CO_2 absorbed by soda lime, so volume change = O_2 consumed)
2. **Without soda lime:** Measures the net gas exchange (CO_2 produced minus O_2 consumed). If $RQ = 1$, the volumes are equal and the fluid drop does not move (CO_2 replaces O_2 exactly). If $RQ < 1$, more O_2 is consumed than CO_2 produced, and the drop moves towards the organism

Volume of CO_2 produced = O_2 uptake (from the set-up WITH soda lime) – net decrease in gas volume (from the set-up WITHOUT soda lime). Then $RQ = CO_2 \text{ produced} \div O_2 \text{ taken up}$.

Key Points

- Germinating seeds have high metabolic rates (actively growing) — ideal for measuring respiration
- Soda lime absorbs CO_2 ; without it, CO_2 replaces some O_2 and the volume change is smaller
- Glass bead control accounts for atmospheric pressure and temperature changes
- $Q_{10} \approx 2$ confirms enzyme-catalysed process (up to optimum temperature, then enzymes denature)

Exam Tip: RQ tells you the respiratory substrate. $RQ = 1.0$: carbohydrate; $RQ \approx 0.7$: lipid; $RQ \approx 0.8$: protein. $RQ > 1$ indicates some anaerobic respiration. In germinating seeds, RQ is typically close to 1 (using stored starch/sugars). During intense exercise, muscles switch from aerobic ($RQ \sim 0.85$ on mixed diet) to partly anaerobic respiration (RQ rises above 1 as CO_2 from lactate buffering adds to aerobic CO_2).

PAG5: Colorimeter or Potometer — Membrane Permeability and Transpiration

Background Theory

PAG5 is about using an **instrument** to turn a biological change into a number. A **colorimeter** shines light of one colour (selected by a filter) through a sample and measures how much is absorbed: the more coloured substance in solution, the higher the **absorbance**. A **potometer** measures the rate at which a leafy shoot takes up water, which is used to estimate the rate of transpiration.

The classic colorimeter practical uses **beetroot**. Beetroot cells store a red-purple pigment (betalain) in the **vacuole**, enclosed by the tonoplast, with the cell-surface membrane outside that. The pigment can only leak out if these membranes are damaged. As temperature rises, phospholipids gain kinetic energy and the bilayer becomes more fluid and leaky; at high temperatures (roughly above 45–50°C) **membrane proteins denature**, leaving gaps, so pigment leaks out rapidly. Organic solvents such as ethanol have a similar effect because they dissolve the lipid bilayer.

Apparatus

- Fresh beetroot, cork borer, scalpel, white tile, ruler
- Water baths (or beakers of water with thermometers) set at a range of temperatures, e.g. 20, 30, 40, 50, 60, 70°C
- Test tubes containing a measured volume (e.g. 5 cm³) of distilled water, stopwatch, forceps
- Colorimeter with a **blue-green** filter (the complementary colour to the red-purple pigment), cuvettes, pipettes
- For the potometer option: bubble potometer, capillary tube with scale, leafy shoot, sink or bowl of water, petroleum jelly, reservoir with tap or syringe

Method (Colorimeter: Temperature and Membrane Permeability)

1. Cut beetroot cylinders of the same diameter with the cork borer and trim them to the same length (e.g. 1 cm) on a white tile.
2. **Rinse the cylinders** in running water (or several changes of distilled water) until no more colour comes out. This removes pigment released from cells that were cut open, so later colour comes only from membrane damage.
3. Put a tube of distilled water in each water bath and allow it to reach the set temperature.
4. Add one cylinder to each tube and leave for a fixed time (e.g. 20 minutes).
5. Remove the cylinders, shake each tube gently to mix, and pour a sample into a cuvette.

6. **Zero the colorimeter** with a cuvette of distilled water, then read the absorbance of each sample. Handle cuvettes by the ridged sides so fingerprints do not scatter light.
7. Repeat at least three times at each temperature and calculate the mean absorbance.

Key Variables

Variable	Detail
Independent	Temperature of the water (°C)
Dependent	Absorbance of the surrounding solution (proportional to the amount of pigment released)
Control variables	Size and surface area of cylinders, volume of water, time in the water bath, colorimeter filter, same beetroot

Method (Potometer: Rate of Water Uptake)

1. Cut a leafy shoot **under water**, with a slanting cut, so air cannot enter the xylem and break the water column.
2. Assemble the potometer under water and fit the shoot so the apparatus is completely water-filled. Seal joints with petroleum jelly so it is airtight and watertight.
3. Remove the apparatus from the water, dry the leaves gently and let the shoot adjust to the conditions for about 5 minutes.
4. Introduce one air bubble into the capillary tube. Record how far it moves in a fixed time.
5. Use the reservoir to push the bubble back to the start, then repeat. Change one factor (e.g. air movement from a fan, light intensity, temperature) and compare rates.

Analysis

- Colorimeter: plot mean absorbance against temperature. Expect a gradual rise at lower temperatures, then a steep rise once membrane proteins denature.
- If the colorimeter is calibrated with pigment solutions of known concentration, a **calibration curve** can convert absorbance into concentration.

- Potometer: volume of water taken up = $\pi r^2 \times \text{distance moved by the bubble}$, where r is the radius of the capillary bore. Rate = volume / time.
- A potometer measures **water uptake**, not transpiration directly: a small amount of the water is used in photosynthesis or to keep cells turgid.

Safety

- Cut beetroot and shoots on a tile, away from the body.
- Water at 60–70°C can scald; use tongs or forceps to move tubes.
- Keep electrical equipment (colorimeter, water baths, fans) away from spills; potometers must be assembled in a sink or bowl.

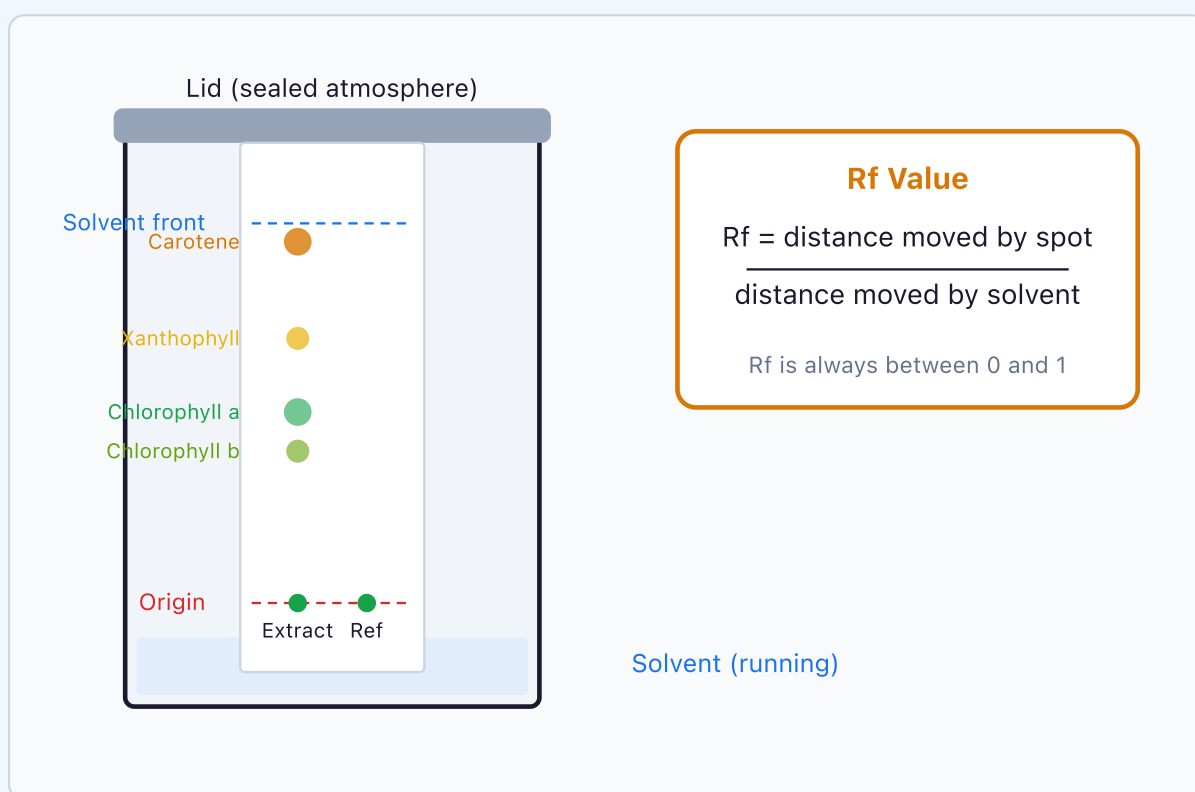
Sources of Error and Improvements

Error	Effect	Improvement
Cylinders not rinsed before the experiment	Pigment from cut cells raises every reading, hiding the effect of temperature	Rinse until the water runs clear before starting
Solution not mixed before sampling	Pigment concentrated near the cylinder; absorbance readings vary	Shake or invert each tube gently before filling the cuvette
Colorimeter not re-zeroed	Drift gives a systematic error in all readings	Zero with distilled water before each set of readings
Air leaks into the potometer or xylem	Bubble moves erratically or stops; rate underestimated	Cut and assemble under water; seal every joint with petroleum jelly
Shoot not given time to adjust	First readings do not reflect the new conditions	Leave 5 minutes after each change before taking readings

Exam Tip: Explain the beetroot results in two stages: at moderate temperatures the phospholipid bilayer becomes more fluid (molecules have more kinetic energy), so permeability rises slowly; at high temperatures the membrane **proteins denature**, creating

large gaps, so absorbance rises steeply. Always say why you rinse the cylinders and why you zero the colorimeter. For potometer questions, remember the formula $\pi r^2 l$ for the volume of water, and state that a potometer measures water uptake rather than water loss.

PAG6: Chromatography — Separating Photosynthetic Pigments



Background Theory

Chromatography separates mixtures of substances based on their **differential solubility** in two phases: a **stationary phase** (the chromatography paper or TLC silica coating) and a **mobile phase** (the solvent that moves up through the stationary phase by capillary action). Each substance in the mixture partitions differently between the two phases based on its relative affinity (solubility) for each. More soluble substances in the mobile phase travel further; substances with greater affinity for the stationary phase are retained near the origin.

Photosynthetic pigments in plant leaves include: **chlorophyll a** (blue-green, the primary photosynthetic pigment), **chlorophyll b** (yellow-green, accessory pigment), **carotene** (orange-yellow, accessory pigment and photoprotective), and **xanthophyll** (yellow, accessory pigment). Each has a different R_f value in a given solvent system.

PAG6 can be met by chromatography **or** electrophoresis. OCR's own example is identifying the **amino acids in a protein** by paper chromatography: the protein is hydrolysed, spots are run alongside known amino acids, and the colourless spots are revealed by spraying with **ninhydrin** (a harmful reagent, used in a fume cupboard with gloves, following CLEAPSS guidance).

Apparatus

- Fresh green leaves (spinach, nettle, or privet), mortar and pestle, clean sharp sand (to aid grinding), propanone (acetone)
- Chromatography paper or thin-layer chromatography (TLC) plate (silica gel on plastic or glass backing)
- Chromatography tank (large beaker or jar) with a tight-fitting lid
- Solvent: petroleum ether : propanone mixture (typically 9:1), or other suitable non-polar/polar solvent mix
- Capillary tube or micropipette (for spotting), pencil, ruler, hairdryer (for drying spots)

Method

1. **Extract pigments:** Grind 5–10 leaves with a pinch of clean sand (abrasive to break cell walls) and $\sim 5 \text{ cm}^3$ of propanone in a mortar and pestle until a dark green solution is obtained. Filter through muslin or filter paper to remove debris.
2. **Prepare the chromatogram:** Draw a faint **pencil line** (the origin line) 2 cm from the bottom of the chromatography paper or TLC plate. Never use pen — ink dissolves in the solvent and interferes with the chromatogram.
3. **Spot the extract:** Using a capillary tube, apply a small, concentrated spot of the pigment extract onto the origin line. Allow to dry completely. Repeat 3–5 times at the same spot to build up concentration (each layer must dry before the next is applied).
4. **Run the chromatogram:** Pour solvent into the tank to a depth of $\sim 1 \text{ cm}$ (the solvent level must be **below** the origin line, or the spots will dissolve directly into the solvent). Place the

paper or TLC plate in the tank. Replace the lid immediately to create a saturated solvent atmosphere (prevents uneven evaporation).

5. Leave undisturbed until the solvent front is ~2 cm from the top of the paper. Remove and **immediately mark the solvent front** with a pencil (it will evaporate quickly).
6. **Measure distances:** From the origin line to the centre of each pigment spot, and from the origin line to the solvent front. Calculate R_f for each pigment.

R_f Value Calculation

R_f = distance moved by the pigment spot / distance moved by the solvent front

R_f values are always between 0 and 1. They are specific to a particular solvent system, temperature, and stationary phase. Compare with published reference values to identify unknown pigments.

Expected Results (Typical Order, Highest to Lowest R_f)

The exact R_f values, and sometimes the order of the middle bands, depend on the solvent and the stationary phase used; carotene always runs furthest in a non-polar solvent.

Pigment	Colour	Approximate R_f	Explanation
Carotene	Orange-yellow	~0.95	Most soluble in the non-polar solvent (highly non-polar hydrocarbon structure); least retained by polar stationary phase
Xanthophyll	Yellow	~0.7	Slightly polar (hydroxyl groups); intermediate solubility
Chlorophyll a	Blue-green	~0.5	Less soluble in the non-polar solvent than carotene; more retained by polar stationary phase
Chlorophyll b	Yellow-green	~0.4	Most polar of the four pigments (additional polar groups); least soluble in non-polar solvent; most

retained near origin

Safety

- Propanone (acetone) and petroleum ether are **highly flammable** — absolutely no naked flames. Work in a well-ventilated area or a fume cupboard.
- Solvent vapours can cause dizziness, headaches, and respiratory irritation — ensure good ventilation; avoid prolonged inhalation.
- Wear safety goggles and gloves. Avoid skin contact with solvents.

Sources of Error and Improvements

Error	Effect	Improvement
Origin line drawn in ink instead of pencil	Ink dissolves and runs with the solvent, contaminating the chromatogram	Always use pencil (graphite is insoluble in organic solvents)
Spot too large, diffuse, or not concentrated enough	Faint, spread-out spots that are hard to measure or identify	Apply small, concentrated spots; allow to dry fully between each application; use 3–5 applications
Solvent level above the origin line when placed in tank	Spots dissolve directly into the solvent instead of being carried up by capillary action — no separation occurs	Ensure solvent depth is well below the origin line (~1 cm depth, origin line at 2 cm from bottom)
Lid not replaced on tank	Solvent evaporates unevenly from the paper, causing an uneven solvent front and inconsistent R_f values	Replace the lid immediately and do not open until complete. Line the tank with solvent-soaked filter paper to saturate the atmosphere
Solvent front not marked immediately on removal	Solvent evaporates rapidly; cannot calculate accurate R_f	Mark the solvent front with pencil within seconds of removing the chromatogram

Exam Tip: R_f values are specific to the solvent system used. If comparing chromatograms, you must confirm that the same solvent, temperature, and stationary phase were used, otherwise R_f values are not comparable. The origin line must be **above the solvent level**. Chlorophyll a and b have different R_f values because they differ in polarity (chlorophyll b has an additional polar aldehyde group that makes it more soluble in the polar stationary phase and less soluble in the non-polar mobile phase). TLC gives better resolution and more reproducible R_f values than paper chromatography.

PAG6: Electrophoresis — Separating DNA Fragments

Background Theory

Gel electrophoresis separates DNA (or protein) fragments by size using an electric field applied across a gel matrix. DNA is **negatively charged** (due to phosphate groups in the sugar-phosphate backbone) and migrates towards the **positive electrode (anode)** when current is applied. The agarose gel acts as a **molecular sieve**: smaller fragments encounter less resistance and move faster/further; larger fragments are retarded by the gel pores and move more slowly. After separation, fragments appear as distinct **bands** at different positions, which can be visualised by staining (e.g. ethidium bromide + UV light, or SYBR Safe stain).

Apparatus

- Pre-prepared agarose gel in an electrophoresis tank (or materials to cast gel: agarose powder, buffer, gel tray, comb)
- DNA samples (pre-cut with restriction enzymes), DNA size ladder (molecular weight markers)
- Micropipettes and tips (for loading), loading dye (adds density so sample sinks into wells; includes tracking dye)
- Electrophoresis tank with electrodes, power supply (typically 100–120 V)
- Buffer solution (TAE or TBE — conducts current and maintains pH)
- Staining solution (SYBR Safe or methylene blue) and/or UV transilluminator

Method

1. If casting: dissolve agarose in buffer, pour into gel tray with comb in place, allow to set.
Remove comb (creates wells).
2. Place the gel in the electrophoresis tank. Add buffer to cover the gel completely.
3. Load samples mixed with loading dye into the wells using a micropipette. Load a **DNA ladder** into one well (provides size reference).
4. Connect the electrodes: negative (black) at the wells end, positive (red) at the far end. DNA migrates from – to +.
5. Run at ~100 V for 30–60 minutes (until the tracking dye has migrated ~75% of the gel length).
6. Turn off the power supply. Remove the gel. Stain if not pre-stained.
7. Visualise bands using UV transilluminator or blue-light box. **Photograph** the gel for analysis.
8. Determine fragment sizes by comparing band positions with the DNA ladder.

Applications

- **DNA profiling:** Compare STR patterns between crime scene samples and suspects
- **Restriction mapping:** Determine the number and sizes of fragments produced by restriction enzyme digestion
- **Checking PCR products:** Confirm that PCR amplified the correct-sized fragment
- **Paternity testing:** Children inherit one allele from each parent at each locus; band sharing can confirm parentage

Safety

- Ethidium bromide is a **mutagen/carcinogen** — wear gloves at all times; dispose of gels and solutions as hazardous waste. Safer alternatives (SYBR Safe, methylene blue) are now commonly used in schools.
- Do not look directly at UV light from the transilluminator — wear UV-protective goggles or use a UV shield.
- Handle the power supply with care — electrical hazard. Never touch the buffer or gel while the power supply is running.

Sources of Error and Improvements

Error	Effect	Improvement
Wells overloaded with DNA	Bands are smeared and indistinct; difficult to determine sizes	Load the recommended volume (typically 5–15 μL); use a micropipette for accuracy
Gel run too fast (voltage too high)	Bands are fuzzy due to heating; poor resolution	Run at recommended voltage ($\sim 100\text{V}$ for standard gels); use a buffer that dissipates heat
DNA ladder not included	Cannot determine fragment sizes; gel is uninterpretable	Always include a DNA size ladder in at least one lane
Air bubbles in wells during loading	Sample is displaced; uneven loading; missing bands	Expel air from pipette tip before loading; practise loading technique

Exam Tip: DNA moves to the **positive** electrode because it is negatively charged (phosphate groups). **Smaller fragments move further** because they pass through gel pores more easily. A DNA ladder allows you to estimate fragment sizes by interpolation. In DNA profiling: identical twins have the same profile. A child shares $\sim 50\%$ of bands with each biological parent. The probability of two unrelated individuals having the same profile is extremely low when 16+ STR loci are tested.

PAG7: Microbiological Techniques

Background Theory

Microbiology practicals require strict **aseptic technique** to prevent contamination of cultures and to protect the experimenter from potentially harmful microorganisms. This practical covers

streaking for single colonies, serial dilution, pour plates, spread plates, and measurement of antimicrobial effectiveness (zones of inhibition).

Key Aseptic Techniques

Technique	Purpose
Work near a lit Bunsen burner (blue flame)	Creates a convection current (updraft) that carries airborne microorganisms away from the work area
Flame the neck of culture bottles/tubes before and after opening	Kills microorganisms on the rim that could contaminate the culture or the environment
Sterilise inoculating loops by heating to red-hot in the flame	Kills all organisms on the loop between each use; prevent cross-contamination between cultures
Allow loops to cool before touching the culture	A red-hot loop would kill the organisms you are trying to transfer and could cause aerosols
Use sterile (pre-packed) Petri dishes and lift the lid only slightly, at an angle, when inoculating	Reduces the chance of airborne microorganisms landing on the agar. Plastic dishes are never flamed — they would melt
Seal plates with two strips of tape (not fully sealed)	Prevents the plate from being accidentally opened, but allows some gas exchange (avoids anaerobic conditions that could promote growth of dangerous obligate anaerobes)
Incubate at $\leq 25^{\circ}\text{C}$ in schools	Prevents growth of potentially pathogenic organisms that thrive at 37°C (human body temperature)
Incubate plates upside down	Prevents condensation dripping from the lid onto the agar surface, which would cause colonies to merge and spread

Serial Dilution and Colony Counting

1. Prepare a series of 10-fold dilutions: transfer 1 cm^3 of the original culture into 9 cm^3 of sterile diluent (total 10 cm^3) $\rightarrow$ this is a 10^{-1} dilution. Repeat for 10^{-2} , 10^{-3} , etc.
2. Plate out 0.1 cm^3 from each dilution onto separate agar plates using a sterile glass spreader. Incubate.
3. Count colonies on plates with 30–300 colonies (statistically reliable range). Calculate the original concentration: $\text{CFU/cm}^3 = (\text{colonies counted}) / (\text{volume plated} \times \text{dilution factor})$.

Safety

- Treat all microbial cultures as potentially pathogenic, even if using known non-pathogenic strains.
- Never open plates after incubation. Autoclave all used plates before disposal.
- Maximum incubation temperature: **$\leq 25^\circ\text{C}$ in schools.**
- Wash hands thoroughly with antibacterial soap after all microbiology work.
- Wipe bench surfaces with disinfectant before and after work.

Exam Tip: Know why each aseptic technique is used. Common exam question: "Why is the lid of the Petri dish only partly lifted when inoculating?" Answer: to minimise the exposure of the sterile agar surface to airborne microorganisms while still allowing access. Also explain why $\leq 25^\circ\text{C}$: to prevent growth of human pathogens that could cause infection if plates were accidentally opened or contaminated.

PAG8: Transport In and Out of Cells — Osmosis and Water Potential

Background Theory

Osmosis is the net movement of water molecules from a region of higher water potential (Ψ) to a region of lower water potential through a **partially permeable membrane**. Plant cells placed in solutions of different concentrations will gain or lose water by osmosis. In **hypotonic** solutions

(higher Ψ than cell sap), cells gain water, become turgid, and increase in mass. In **hypertonic** solutions (lower Ψ than cell sap), cells lose water, become flaccid (or plasmolysed in extreme cases), and decrease in mass. At the **isotonic point** (equal Ψ), there is no net movement of water and no change in mass.

Apparatus

- Potato tubers (or other firm plant tissue e.g. beetroot, sweet potato), cork borer, scalpel, ruler
- Sucrose solutions at a range of concentrations (e.g. 0.0, 0.2, 0.4, 0.6, 0.8, 1.0 mol dm⁻³), prepared by serial dilution from a stock solution
- Distilled water (0.0 M), boiling tubes or beakers, balance (accurate to 0.01 g)
- Paper towels (for blotting), forceps, labels, stopwatch

Method

1. Use a cork borer to cut **uniform potato cylinders**. Trim each to the same length (e.g. 3 cm) using a ruler and scalpel. This ensures equal surface area for osmosis.
2. Blot each cylinder gently with paper towel to remove surface moisture. **Weigh each cylinder** on a balance (to 0.01 g) and record the initial mass.
3. Place one cylinder in each boiling tube containing 20 cm³ of sucrose solution (one tube per concentration). Ensure the cylinders are fully submerged.
4. Leave for a fixed time (e.g. **30–60 minutes**) at constant room temperature.
5. Remove each cylinder with forceps. **Blot gently** (same technique each time: roll once along a paper towel) to remove surface moisture. Re-weigh and record the final mass.
6. Calculate the **percentage change in mass**: % change = ((final mass – initial mass) / initial mass) x 100.
7. Repeat the entire procedure at least **3 times** per concentration. Calculate the mean % change.

Key Variables

Variable	Detail
Independent	Sucrose concentration (0.0–1.0 mol dm ⁻³)

Dependent	Percentage change in mass of potato cylinder
Control variables	Initial cylinder size (same cork borer, same length), volume of solution (20 cm ³), temperature, immersion time, blotting technique, same potato variety

Analysis

- Plot **mean % change in mass (y-axis)** against **sucrose concentration (x-axis)**.
- Expected: positive % change (mass gain) in low concentrations (hypotonic — water enters by osmosis); negative % change (mass loss) in high concentrations (hypertonic — water leaves by osmosis).
- The point where the line crosses the x-axis (% change = 0) indicates the **isotonic concentration** — the solution with the same water potential as the potato cell sap. At this point, no net osmosis occurs.
- Use percentage change rather than absolute change to account for differences in starting mass between cylinders.

Safety

- Sharp cutting tools (scalpel, cork borer) — cut on a white tile, away from the body.
- Sucrose solutions are non-hazardous. Mop up spills to prevent slipping.

Sources of Error and Improvements

Error	Effect	Improvement
Inconsistent cylinder sizes	Different surface area:volume ratios; different rates of osmosis	Use the same cork borer and cut all cylinders to the exact same length with a ruler
Inconsistent blotting technique	Some cylinders have more surface water removed than others; affects mass measurement	Standardise: roll each cylinder once along a dry paper towel using the same pressure

Different immersion times for different tubes	Confounding variable: longer immersion allows more osmosis	Start all tubes simultaneously or stagger starts at 1-minute intervals and adjust removal accordingly
Temperature variation	Affects rate of osmosis (kinetic energy of water molecules)	Keep all tubes at the same location/temperature; record temperature
Using absolute mass change instead of percentage change	Cannot compare cylinders of different starting masses fairly	Always calculate and plot % change in mass

Exam Tip: Always use **percentage change** in mass, not absolute change, to standardise for different starting masses. The isotonic point (where the graph crosses the x-axis) estimates the water potential of the potato cell sap. In a hypotonic solution, cells gain water by osmosis (positive % change); in a hypertonic solution, cells lose water (negative % change). If asked about plasmolysis: this occurs when the cell membrane pulls away from the cell wall as the cell loses water in a very concentrated solution — the cell wall is fully permeable, but the cell membrane is not, so it shrinks within the rigid wall.

PAG9: Qualitative Testing for Biological Molecules

Background Theory

Biochemical tests show whether a particular type of molecule is present in a sample such as a food, a plant extract or a body fluid. The OCR B specification links these tests to medicine and diagnosis: you need the method and interpretation of **Benedict's test**, the **biuret test**, the **iodine test** for starch and **colorimetry**, and you should know that **reagent test strips** and **biosensors** can detect and measure sugars and proteins (for example glucose in urine or blood).

A qualitative test only says whether a substance is present. By standardising volumes and heating time, and comparing colours with known standards (or using a colorimeter), Benedict's

test can be made **semi-quantitative**.

Apparatus

- Test tubes and rack, pipettes or syringes for measuring volumes, spotting tile
- Benedict's reagent, biuret reagent (or dilute sodium hydroxide and dilute copper(II) sulfate), iodine in potassium iodide solution, ethanol
- Dilute hydrochloric acid and sodium hydrogencarbonate (for non-reducing sugars)
- Water bath at about 80–100°C, glucose solutions of known concentration, colorimeter, glucose test strips

Method

Test for	Method	Positive result
Reducing sugar (e.g. glucose, fructose, maltose, lactose)	Add an equal volume of Benedict's reagent to the sample and heat in a water bath at 80–100°C for a few minutes	Blue → green → yellow → orange → brick-red precipitate , depending on how much sugar is present
Non-reducing sugar (e.g. sucrose)	First show the Benedict's test is negative. Then boil a fresh sample with dilute HCl to hydrolyse glycosidic bonds, neutralise with sodium hydrogencarbonate, and repeat the Benedict's test	Now gives a coloured precipitate, because hydrolysis released reducing monosaccharides
Starch	Add a few drops of iodine in potassium iodide solution	Orange-brown → blue-black
Protein	Add biuret reagent (or sodium hydroxide, then a few drops of dilute copper(II) sulfate); no heating	Blue → lilac / purple (copper ions react with peptide bonds)
Lipid	Shake the sample with ethanol, then pour the ethanol into water	White cloudy emulsion

Semi-Quantitative Benedict's Test with a Colorimeter

1. Make a dilution series of glucose solutions of known concentration from a stock solution.
2. Carry out Benedict's test on each using the same volumes, temperature and heating time.
3. Filter off (or centrifuge down) the precipitate. Zero the colorimeter with distilled water, then measure the absorbance of the remaining blue solution using a **red** filter.
4. More glucose uses up more copper(II) ions, so less blue remains and absorbance is **lower**.
Plot a calibration curve of absorbance against glucose concentration.
5. Test the unknown sample the same way and read its glucose concentration off the calibration curve.

Safety

- Wear eye protection: sodium hydroxide in biuret reagent is an irritant/corrosive to eyes, and hydrochloric acid is an irritant.
- Ethanol is highly flammable — no naked flames nearby; use an electric water bath.
- Point boiling tubes away from people when heating.
- Do not use real body fluids; use simulated urine or blood samples.

Sources of Error and Improvements

Error	Effect	Improvement
Judging colour by eye	Subjective; hard to compare similar colours	Use a colorimeter and a calibration curve, or compare against colour standards
Different volumes or heating times	Colour depends on the procedure, not only on sugar concentration	Measure volumes with syringes or pipettes; heat all tubes together for the same time
Not neutralising after acid hydrolysis	Benedict's reagent needs alkaline conditions, so a false negative can result	Add sodium hydrogencarbonate until fizzing stops (check with indicator paper)

Benedict's reagent
not in excess

Very high sugar concentrations
all look the same

Use excess reagent or dilute the
sample first

Exam Tip: State the starting colour as well as the end colour (e.g. "blue to brick-red", not just "red"). For non-reducing sugars, give all the steps: negative first test, acid hydrolysis with heat, neutralise, retest. In colorimetry with Benedict's, remember the direction: more sugar means **less** blue copper(II) left, so lower absorbance of the filtrate. Test strips and biosensors are quicker and more specific than Benedict's, because they use an enzyme (e.g. glucose oxidase) that reacts only with glucose.

PAG10: Investigation Using a Data Logger or Computer Modelling

Background Theory

PAG10 is about using ICT to collect or process data. A **data logger** is a sensor connected to a recorder or computer that takes readings automatically at set intervals (for example heart rate, ECG, temperature, oxygen or carbon dioxide concentration, pH). **Computer modelling** uses software to represent a biological system: the OCR example is using a molecular viewer such as **RasMol** (or Jmol) to explore the structure of DNA. In the OCR B specification, PAG10 is linked to heart rate and the ECG, the effect of exercise, lung function, aerobic fitness and measuring body temperature.

Apparatus

- Data logger with suitable sensors, e.g. a heart rate monitor (chest strap or finger/ear clip), ECG electrodes, temperature probe, oxygen or CO₂ sensor
- Computer or tablet with logging and graphing software
- Step or exercise bike, stopwatch and metronome (for exercise investigations)

- For modelling: a molecular viewer (RasMol or Jmol) and a structure file of DNA from a protein/nucleic acid database

Method A: Data Logging Heart Rate During Exercise and Recovery

1. Fit the heart rate sensor and check it gives a steady reading. Set the sampling interval (e.g. one reading every 5 s).
2. Record the resting heart rate for 3–5 minutes while the participant sits still.
3. Start standardised exercise (e.g. stepping in time with a metronome for 3 minutes) while the logger keeps recording.
4. Continue logging during recovery until the heart rate is back to the resting value.
5. Use the software to plot heart rate against time, and to find the resting mean, the peak value and the recovery time. Repeat for other participants or exercise intensities.

Method B: Modelling DNA Structure with RasMol

1. Open a DNA structure file in the viewer. Display it in different formats (wireframe, ball-and-stick, space-filling) and rotate it.
2. Colour the model by chain to see the **two antiparallel strands**, and by component to pick out the sugar-phosphate backbone and the bases.
3. Identify complementary base pairs (A–T with two hydrogen bonds, G–C with three) and the major and minor grooves.
4. Use the measuring tools to find distances, e.g. the width of the helix (about 2 nm) and the length of one complete turn (about 3.4 nm, roughly 10 base pairs).

Analysis

- From an ECG trace: the **P wave** is atrial depolarisation, the **QRS complex** is ventricular depolarisation, and the **T wave** is ventricular repolarisation.
- Heart rate (beats min^{-1}) = $60 \div \text{time between two R peaks (s)}$.
- Cardiac output = heart rate $\times$ stroke volume.
- Recovery time is a useful measure of aerobic fitness: fitter people usually return to resting heart rate sooner.

Advantages and Limitations of Data Loggers

Advantages	Limitations
Readings taken automatically at precise, frequent intervals, including during exercise when manual counting is hard	Sensors must be calibrated and correctly fitted, or every reading is wrong (systematic error)
Removes human reaction-time and counting errors	Movement or poor contact can cause spikes and gaps in the data
Can record several variables at once and over long periods	Large data sets need sensible processing; averaging can hide short changes
Software plots graphs and calculates values quickly	Cost and availability of sensors

Safety

- Screen participants before exercise (asthma, heart conditions, injuries); stop if they feel unwell. Participants must volunteer and can withdraw.
- Keep sensors and cables away from water and trip hazards.
- Clean reusable electrodes and sensors between users.

Exam Tip: If asked why a data logger is better than taking a pulse by hand, give a specific reason: it records **continuously at a set interval**, so the peak heart rate and the exact recovery time are not missed, and it removes counting errors. For ECG questions, measure the R–R interval carefully using the grid scale and convert to beats per minute with $60 \div \text{interval}$. For DNA models, link what you see to function: complementary base pairing allows accurate replication, and the two strands are antiparallel.

PAG11: Animal Responses — Heart Rate and Ventilation

Background Theory

During exercise, skeletal muscles require more ATP, increasing the rate of aerobic respiration. This raises O₂ demand and CO₂ production. The body responds by increasing both **heart rate** (to deliver more oxygenated blood to muscles) and **ventilation rate** (to increase gas exchange in the lungs). These responses are coordinated by chemoreceptors (detecting blood CO₂/pH changes) and the cardiovascular and respiratory centres in the medulla oblongata.

This OCR B practical investigates ventilation and heart rate responses, and matches OCR's own PAG11 example (the effect of exercise on pulse rate). Heart rate can be measured by pulse palpation (radial or carotid artery) or with a heart rate monitor; tidal volume and breathing rate can be measured with a spirometer.

Method Summary

1. Screen participants for contraindications (asthma, cardiac conditions).
2. Measure **resting heart rate** (count pulse for 60 s) and **resting breathing rate** (count breaths for 60 s). Measure **tidal volume** with a spirometer (a peak flow meter measures peak expiratory flow rate, not tidal volume).
3. Subject performs standardised exercise (step-ups at 30/min for 5 min, or cycle ergometer at set resistance).
4. Measure heart rate, breathing rate, and tidal volume **immediately after exercise** and at 1-minute intervals during recovery.
5. Calculate **minute ventilation** = tidal volume x breathing rate.
6. Plot all measurements against time. Determine recovery time (time to return to resting values).

Key Physiology

- During exercise: CO₂ rises in blood → detected by central chemoreceptors (medulla) and peripheral chemoreceptors (carotid/aortic bodies) → increased sympathetic stimulation → increased heart rate (via SA node) + increased breathing rate and depth (via intercostal muscles and diaphragm)
- Adrenaline (released from adrenal medulla during exercise) also increases heart rate, cardiac output, and blood redistribution to skeletal muscles
- Cardiac output = stroke volume x heart rate. Both increase during exercise
- Fitter individuals: lower resting heart rate, faster recovery time, larger stroke volume

Exam Tip: Minute ventilation = tidal volume x breathing rate. Cardiac output = stroke volume x heart rate. The primary stimulus for increased ventilation during exercise is rising CO₂ (not falling O₂). CO₂ dissolves in blood plasma to form carbonic acid ($\text{H}_2\text{CO}_3 \rightarrow \text{H}^+ + \text{HCO}_3^-$), lowering blood pH. Chemoreceptors detect the pH change and increase ventilation. This is negative feedback maintaining blood gas homeostasis.

PAG11: Plant Responses — Tropisms and Germination

Background Theory

Plants respond to directional stimuli through **tropisms** — growth responses mediated by the hormone **auxin (IAA, indole-3-acetic acid)**. Auxin is produced in the shoot tip and redistributes in response to light (phototropism) and gravity (gravitropism). In shoots, auxin **stimulates cell elongation** (the shaded/lower side elongates more, bending the shoot towards light or away from gravity). In roots, the same concentration of auxin **inhibits cell elongation** (the lower side grows less, bending the root towards gravity). This differential sensitivity is due to different optimal auxin concentrations for shoot vs root cells.

PAG11 can be met with either plant or animal responses. Auxin and tropisms are not named in the H422 content, but a phototropism investigation is a convenient plant option. The H422 content does require the role of **gibberellin** as a cell-signalling molecule in seed germination, investigations into the factors affecting germination, and the role of phytochrome in the timing of flowering (module 4.4.1).

Key Investigation: Auxin and Phototropism

1. Grow oat coleoptiles or cress seedlings uniformly in darkness for 3–5 days.
2. Expose seedlings to **unidirectional light** from one side (use a box with a single hole).
3. After 24–48 hours, measure the angle and direction of curvature.
4. **Controls:** (a) seedlings in darkness (no light stimulus — should grow straight up), (b) seedlings with uniform light from all sides (no directional stimulus), (c) seedlings on a

clinostat (eliminates gravity as a variable), (d) seedlings with the tip removed (no auxin production — should not bend).

Investigating Gibberellin and Germination

GA₃ stimulates the aleurone layer to produce alpha-amylase, which digests starch to maltose, providing energy for the embryo. De-embryonated half-seeds treated with GA₃ show starch digestion (tested with iodine: no blue-black colour if starch has been digested), proving that GA₃ (not the embryo directly) triggers amylase gene expression.

Sources of Error and Improvements

Error	Effect	Improvement
Light leaking into the box from gaps	Not truly unidirectional light; weak or inconsistent phototropism	Seal the box with black tape; use blackout material; work in a dark room
Seedlings at different developmental stages	Different sensitivity to auxin; variable responses	Germinate all seeds at the same time; select seedlings of uniform size
Gravity confounding the light response	Seedlings respond to both gravity and light simultaneously	Use a clinostat to neutralise gravity; compare with and without clinostat
Measuring angle by eye	Subjective and inaccurate	Photograph seedlings against a reference grid; use image analysis software or a protractor

Exam Tip: Auxin causes **different effects in shoots vs roots** at the same concentration. In shoots: auxin promotes cell elongation → the shaded side elongates more → shoot bends towards light (positive phototropism). In roots: the same concentration inhibits elongation → the lower side (where auxin accumulates due to gravity) grows less → root bends downwards (positive gravitropism). The clinostat control is essential to prove that the

response is truly due to gravity (by removing it as a directional stimulus through continuous slow rotation).

PAG12: Research Skills — Planning an Investigation into Yeast Respiration

Background Theory

PAG12 assesses **research skills** rather than one technique. You must apply investigative approaches, use online and offline sources to research a question, and **correctly cite** the sources you used. OCR's example activity is an investigation into the rate of respiration of the yeast *Saccharomyces cerevisiae*. Yeast is a facultative anaerobe: it respire aerobically when oxygen is available and carries out alcoholic fermentation (producing ethanol and CO₂) when it is not. Either way it releases CO₂, so the rate of CO₂ production can be used to measure its rate of respiration.

Stages of a Research-Based Investigation

1. **Research the background:** use textbooks, journals and reliable websites to find out what is already known, e.g. which sugars yeast can use and which enzymes it needs.
2. **Choose a question and hypothesis:** e.g. "the rate of CO₂ production by yeast depends on the respiratory substrate" or "rate increases with temperature up to an optimum".
3. **Plan the method:** identify the independent, dependent and control variables, choose a suitable range and number of repeats, and write a risk assessment.
4. **Carry out a trial run** to check the method works and the range is sensible, then collect the full data.
5. **Analyse and evaluate:** process the data, compare results with your research, and identify limitations.
6. **Reference your sources** in a consistent style.

Apparatus (Example Investigation)

- Dried or fresh yeast, solutions of different sugars (e.g. glucose, sucrose, lactose) at the same concentration, or one sugar at a range of concentrations
- Boiling tubes or conical flasks, delivery tube and gas syringe (or an upturned measuring cylinder in water)
- Thermostatically controlled water bath, stopwatch, measuring cylinders or syringes

Method (Example Investigation)

1. Mix a fixed mass of yeast with a fixed volume of warm water and leave it to activate in the water bath (e.g. 35°C).
2. Warm the substrate solution to the same temperature, then add a measured volume to the yeast suspension and connect the gas syringe immediately.
3. Leave a few minutes for the system to equilibrate, then record the volume of CO₂ collected every minute for about 10 minutes.
4. Repeat with each substrate (or concentration), including a control with no sugar, and carry out at least three repeats.

Key Variables

Variable	Detail
Independent	Type of respiratory substrate (or its concentration, or temperature)
Dependent	Rate of CO ₂ production (cm ³ min ⁻¹)
Control variables	Mass and type of yeast, volume and concentration of substrate, temperature, pH, time allowed to activate

Judging and Citing Sources

Question to ask	Why it matters
Who wrote it, and are they an expert?	Universities, learned societies and peer-reviewed journals are more reliable than anonymous pages

Is it recent?	Scientific understanding and methods change over time
Is there bias?	A company selling a product may present only favourable evidence
Do other sources agree?	Agreement between independent sources increases confidence

Cite every source you use. A typical reference gives the author(s), year, title, publisher or journal (with volume and page numbers), and for a website the full URL and the date you accessed it. Put a short citation in the text where you use the information, and the full reference in a list at the end.

Safety

- Yeast can trigger allergies; avoid creating dust from dried yeast.
- Take care with hot water baths and glassware; clean up spills to avoid slipping.

Sources of Error and Improvements

Error	Effect	Improvement
Gas escapes before the syringe is connected	Rate underestimated, especially early on	Connect straight after mixing; use an equilibration period and then start timing
Some CO ₂ dissolves in the solution	Collected volume is less than the volume produced	Collect gas in a gas syringe rather than over water; keep volumes the same in every run so any error is consistent
Temperature varies between runs	Confounds the effect of the substrate	Use a thermostatically controlled water bath and record the temperature
Only one or two sources used for research	Hypothesis and method may rest on unreliable information	Use several independent, reputable sources and note where they disagree

Exam Tip: Questions on research skills often ask you to evaluate a source or justify a method. Say **why** a source is reliable (peer-reviewed, recent, expert author, no conflict of

interest), not just that it is. For the yeast investigation, remember that glucose is used fastest, sucrose must first be hydrolysed by the yeast's own enzyme, and yeast cannot use lactose because it lacks the enzyme to break it down. Rate is the gradient of the volume–time graph, so use the linear part of the curve.